EXAMPLE 6-4

Boost Converter Design 1

Design a boost converter that will have an output of 30 V from a 12-V source. Design for continuous inductor current and an output ripple voltage of less than one percent. The load is a resistance of 50 Ω. Assume ideal components for this design.

■ Solution

First, determine the duty ratio from Eq. (6-27),

$$D = 1 - \frac{V_s}{V_o} = 1 - \frac{12}{30} = 0.6$$

If the switching frequency is selected at 25 kHz to be above the audio range, then the minimum inductance for continuous current is determined from Eq. (6-32).

$$L_{\min} = \frac{D(1-D)^2(R)}{2f} = \frac{0.6(1-0.6)^2(50)}{2(25{,}000)} = 96\ \mu\text{H}$$

To provide a margin to ensure continuous current, let $L = 120\ \mu$H. Note that L and f are selected somewhat arbitrarily and that other combinations will also give continuous current.

Using Eqs. (6-28) and (6-25),

$$I_L = \frac{V_s}{(1-D)^2(R)} = \frac{12}{(1-0.6)^2(50)} = 1.5\ \text{A}$$

$$\frac{\Delta i_L}{2} = \frac{V_s DT}{2L} = \frac{(12)(0.6)}{(2)(120)(10)^{-6}(25{,}000)} = 1.2\text{ A}$$

$$I_{\max} = 1.5 + 1.2 = 2.7\text{ A}$$

$$I_{\min} = 1.5 - 1.2 = 0.3\text{ A}$$

The minimum capacitance required to limit the output ripple voltage to 1 percent is determined from Eq. (6-35).

$$C \geq \frac{D}{R(\Delta V_o / V_o) f} = \frac{0.6}{(50)(0.01)(25{,}000)} = 48\ \mu\text{F}$$

EXAMPLE 6-5

Boost Converter Design 2

A boost converter is required to have an output voltage of 8 V and supply a load current of 1 A. The input voltage varies from 2.7 to 4.2 V. A control circuit adjusts the duty ratio to keep the output voltage constant. Select the switching frequency. Determine a value for the inductor such that the variation in inductor current is no more than 40 percent of the average inductor current for all operating conditions. Determine a value of an ideal capacitor such that the output voltage ripple is no more than 2 percent. Determine the maximum capacitor equivalent series resistance for a 2 percent ripple.

■ Solution

Somewhat arbitrarily, choose 200 kHz for the switching frequency. The circuit must be analyzed for both input voltage extremes to determine the worst-case condition. For $V_s = 2.7$ V, the duty ratio is determined from Eq. (6-27).

$$D = 1 - \frac{V_s}{V_o} = 1 - \frac{2.7}{8} = 0.663$$

Average inductor current is determined from Eq. (6-28).

$$I_L = \frac{V_o I_o}{V_s} = \frac{8(1)}{2.7} = 2.96\text{ A}$$

The variation in inductor current to meet the 40 percent specification is then $\Delta i_L = 0.4(2.96) = 1.19$ A. The inductance is then determined from Eq. (6-33).

$$L = \frac{V_s D}{\Delta i_L f} = \frac{2.7(0.663)}{1.19(200{,}000)} = 7.5\ \mu\text{H}$$

Repeating the calculations for $V_s = 4.2$V,

$$D = 1 - \frac{V_s}{V_o} = 1 - \frac{4.2}{8} = 0.475$$

$$I_L = \frac{V_o I_o}{V_s} = \frac{8(1)}{4.2} = 1.90\text{ A}$$

The variation in inductor current for this case is $\Delta i_L = 0.4(1.90) = 0.762$ A, and

$$L = \frac{V_s D}{\Delta i_L f} = \frac{4.2(0.475)}{0.762(200{,}000)} = 13.1\ \mu\text{H}$$

The inductor must be 13.1 μH to satisfy the specifications for the total range of input voltages.

Equation (6-35), using the maximum value of D, gives the minimum capacitance as

$$C = \frac{D}{R(\Delta V_o/V_o)f} = \frac{D}{(V_o/I_o)(\Delta V_o/V_o)f} = \frac{0.663}{(8/1)(0.02)(200{,}000)} = 20.7\ \mu\text{F}$$

The maximum ESR is determined from Eq. (6-36), using the maximum peak-to-peak variation in capacitor current. The peak-to-peak variation in capacitor current is the same as maximum inductor current. The average inductor current varies from 2.96 A at $V_s = 2.7$ V to 1.90 A at $V_s = 4.2$ V. The variation in inductor current is 0.762 A for $V_s = 4.2$ A, but it must be recalculated for $V_s = 2.7$ V using the 13.1-μH value selected, yielding

$$\Delta i_L = \frac{V_s D}{Lf} = \frac{2.7(0.663)}{13.1(10)^{-6}(200{,}000)} = 0.683\ \text{A}$$

Maximum inductor current for each case is then computed as

$$I_{L,\max,2.7\text{V}} = I_L + \frac{\Delta i_L}{2} = 2.96 + \frac{0.683}{2} = 3.30\ \text{A}$$

$$I_{L,\max,4.2\text{V}} = I_L + \frac{\Delta i_L}{2} = 1.90 + \frac{0.762}{2} = 2.28\ \text{A}$$

This shows that the largest peak-to-peak current variation in the capacitor will be 3.30 A. The output voltage ripple due to the capacitor ESR must be no more than (0.02)(8) = 0.16 V. Using Eq. (6-36),

$$\Delta V_{o,\text{ESR}} = \Delta i_C r_C = I_{L,\max} r_C = 3.3 r_C = 0.16\ \text{V}$$

which gives

$$r_C = \frac{0.16\ \text{V}}{3.3\ \text{A}} = 48\ \text{m}\Omega$$

In practice, a capacitor that has an ESR of 48 mΩ or less could have a capacitance value much larger than the 20.7 μF calculated.

Inductor Resistance

Inductors should be designed to have small resistance to minimize power loss and maximize efficiency. The existence of a small inductor resistance does not substantially change the analysis of the buck converter as presented previously in this chapter. However, inductor resistance affects performance of the boost converter, especially at high duty ratios.

For the boost converter, recall that the output voltage for the ideal case is

$$V_o = \frac{V_s}{1 - D} \tag{6-37}$$

To investigate the effect of inductor resistance on the output voltage, assume that the inductor current is approximately constant. The source current is the same as the inductor current, and average diode current is the same as average load current. The power supplied by the source must be the same as the power absorbed by the load and the inductor resistance, neglecting other losses.

$$\begin{aligned} P_s &= P_o + P_{r_L} \\ V_s I_L &= V_o I_D + I_L^2 r_L \end{aligned} \tag{6-38}$$

where r_L is the series resistance of the inductor. The diode current is equal to the inductor current when the switch is off and is zero when the switch is on. Therefore, the average diode current is

$$I_D = I_L(1 - D) \tag{6-39}$$

Substituting for I_D into Eq. (6-38),

$$V_s I_L = V_o I_L(1 - D) + I_L^2 r_L$$

which becomes

$$V_s = V_o(1 - D) + I_L r_L \tag{6-40}$$

In terms of V_o from Eq. (6-39), I_L is

$$I_L = \frac{I_D}{1 - D} = \frac{V_o/R}{1 - D} \tag{6-41}$$

Substituting for I_L into Eq. (6-40),

$$V_s = \frac{V_o r_L}{R(1 - D)} + V_o(1 - D)$$

Solving for V_o,

$$\boxed{V_o = \left(\frac{V_s}{1 - D}\right)\left(\frac{1}{1 + r_L/[R(1 - D)^2]}\right)} \tag{6-42}$$

The preceding equation is similar to that for an ideal converter but includes a correction factor to account for the inductor resistance. Figure 6-10*a* shows the output voltage of the boost converter with and without inductor resistance.

The inductor resistance also has an effect on the power efficiency of converters. Efficiency is the ratio of output power to output power plus losses. For the boost converter

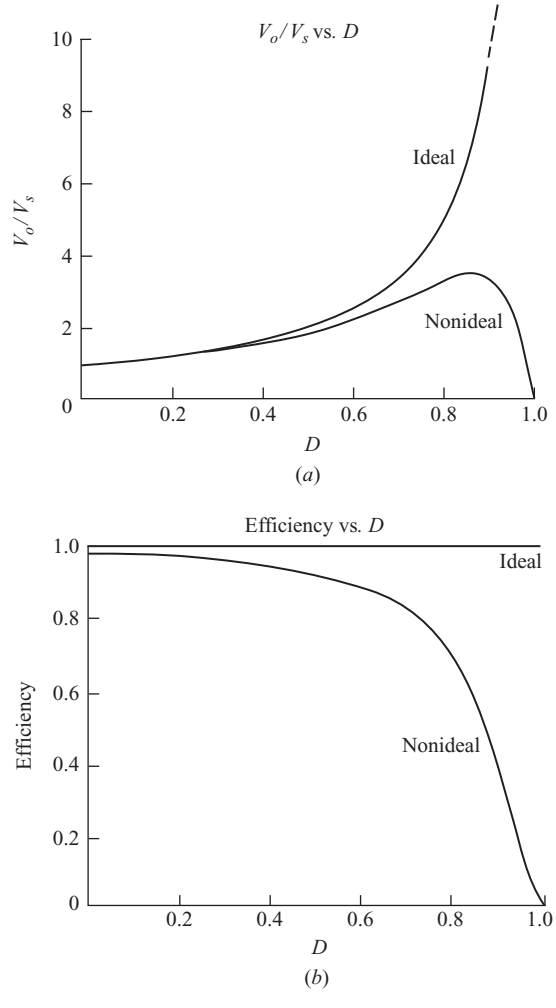


Figure 6-10 Boost converter for a nonideal inductor. (*a*) Output voltage; (*b*) Boost converter efficiency.

$$\eta = \frac{P_o}{P_o + P_{\text{loss}}} = \frac{V_o^2/R}{V_o^2/R + I_L^2 r_L} \tag{6-43}$$

Using Eq. (6-41) for I_L,

$$\eta = \frac{V_o^2/R}{V_o^2/R + (V_o/R)^2/(1-D)r_L} = \frac{1}{1 + r_L[R(1-D)^2]} \tag{6-44}$$

As the duty ratio increases, the efficiency of the boost converter decreases, as indicated in Fig. 6-10*b*.